

# MATH 351 second midterm checklist

## General

- I have read all the assigned sections of the textbook.
- I have read all the feedback on Blackboard for my homework assignments.
- I have studied all the solution sets for homework assignments on Blackboard.
- I have studied all the solution sets for quizzes on Blackboard.

## New topics since first midterm exam

### Conditional probability

- I know the definition of conditional probability: for events  $A$  and  $B$  where  $P(B) > 0$ , the probability of  $A$  given  $B$  is  $P(A|B) = P(A \cap B) / P(B)$ .
- I know that to say events  $A$  and  $B$  are independent (where  $P(B) > 0$ ) is equivalent to saying  $P(A | B) = P(A)$ .
- I can calculate conditional probabilities using the definition.
- I can calculate conditional probabilities using the “reducing the sample space” method.
- I can reverse the definition of conditional probability and use it as a multiplication rule for calculating  $P(A \cap B)$  when  $A$  and  $B$  are not necessarily independent:  $P(A \cap B) = P(A) P(B|A)$ .
- I know that a partition of a sample space is a list of events  $B_1, \dots, B_n$  that are pairwise disjoint (for any  $i$  and  $j$  with  $i \neq j$ ,  $B_i \cap B_j = \emptyset$ ) and whose union is the whole sample space ( $B_1 \cup \dots \cup B_n = \Omega$ ).
- I know that for any event  $B$ , the events  $B$  and  $B^c$  form a partition of the sample space.
- I can use the law of total probability to calculate  $P(A)$  given a partition  $B_1, \dots, B_n$  of a sample space:  $P(A) = P(B_1) P(A|B_1) + \dots + P(B_n) P(A|B_n)$ . I know that with this problem-solving technique, we think of our chance experiment as a “two-stage experiment.”
- I know sophisticated examples of the two-stage-experiment technique including cases where  $A$  is related to a random variable  $X$  and we know the distribution of  $X$  conditional on the occurrence of each event  $B_i$  in our partition. An example is the random variable  $X$  where if event  $B_i$  occurs, we flip  $i$  coins and count the number  $X$  of heads. In this example, we know the distribution of  $X$  conditional on  $B_i$  ( $X|B_i \sim \text{binomial}(i, 1/2)$ ) and can use the law of total probability to calculate unconditional probabilities related to  $X$ .
- I know the Bayes'-formula method for calculating conditional probabilities. I can explain the method as in Durrett, Quiz 7, and various homework questions.
- I know what the joint distribution of two discrete random variables is. I can calculate the joint distribution and present it in a table as in Durrett and homework.
- I can calculate the marginal distributions (separate distributions for  $X$  and  $Y$ ) given the joint distribution of discrete random variables  $X$  and  $Y$ .
- I know what it means for two random variables to be independent. I can decide whether two discrete random variables are independent using their joint distribution.

- I can use the joint distribution of discrete random variables  $X$  and  $Y$  to calculate conditional probabilities such as  $P(X = i \mid Y = j)$  and conditional expectations such as  $E[X \mid Y = j]$ .

## Continuous random variables

- I know the informal definition of a continuous random variable as a random variable that can take any value on the real number line or perhaps any value in an interval of the number line.
- I know that for a continuous random variable  $X$ , we consider events of the form  $(a \leq X \leq b)$ .
- I know that a continuous random variable  $X$  has an associated density function  $f$  such that  $P(a \leq X \leq b) = \int_a^b f(x)dx$ .
- I know that a density function  $f$  must satisfy  $\int_{-\infty}^{\infty} f(x)dx = 1$  (improper integral) and that  $f$  must have non-negative values.
- I know that for any random variable  $X$ , the distribution function  $F$  for  $X$  is defined by  $F(x) = P(X \leq x)$ . I know that distribution functions have limit 0 as  $x \rightarrow -\infty$ , have limit 1 as  $x \rightarrow +\infty$ , and are non-decreasing.
- I know that for a continuous random variable with density function  $f$ ,  $F(x) = \int_{-\infty}^x f(x)dx$  (improper integral). I know that in this case, the distribution function  $F$  is continuous and  $F' = f$ . (To make that statement precise requires care, for example, at points where  $F$  is not differentiable; we will not consider those details.)
- I know formulas for the density function, the distribution function, the expected value, and the variance of a random variable with the exponential( $\lambda$ ) distribution.
- I know how to pick the parameter  $\lambda$  to match an exponential( $\lambda$ ) random variable  $X$  to quantitative data by matching  $E[X] = \frac{1}{\lambda}$  with the mean of the data values. (This way of choosing parameters in a probability model for data is sometimes called “the method of moments.”)
- I know formulas for the density function, the distribution function, the expected value, and the variance of a random variable that is uniformly distributed on the interval  $[a, b]$  (for real numbers  $a$  and  $b$ ).

## Topics from first midterm exam

You should still know what you learned before the first midterm exam! In this class, we continually build on what came before.

## Formalism of probability

- I know the formalism for describing chance experiments: outcomes, sample space, events.
- I know the operations of the algebra of events and can use these operations to express complex events in terms of simpler events. For example, the event “at least one of  $A$ ,  $B$ ,  $C$  occurs” means “ $A \cup B \cup C$ ”.
- I know that we assign probabilities to events.
- I know the rules of probability, including the basic rules and derived rules such as  $P(A^c) = 1 - P(A)$  and  $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ .
- I know the formal definition of “ $A$  and  $B$  are independent.”

- I know what “A and B are independent” means informally.
- I know when one adds probabilities of events and when one multiplies probabilities of events.

## Random variables

- I know the formal definition of a random variable.
- I know the formal definition of the distribution of a random variable.
- I can calculate the distribution of a random variable given a description chance experiment and a random variable for that experiment.
- I can use the “ $\sim$ ” notation to indicate that a random variable has one of our named distributions (binomial, Poisson, geometric so far).
- I know the formal definition of the expected value of a random variable.
- I know how to interpret expected value informally, as
- I can calculate the expected value of a random variable given its distribution.
- I know the algebraic properties of expected value.
- I know what the moments of a random variable are, and I can calculate them given the distribution of a random variable using the Law of the Unconscious Statistician (LOTUS).
- I know how to calculate the variance of a random variable  $X$  using  $E[X^2]$  and  $E[X]$ .
- I know how to calculate  $\sigma(X)$  using  $\text{Var}(X)$  for a random variable  $X$ .
- I know how to match a random variable described in terms of a chance experiment with the story of the binomial distribution.
- I know the formula for the binomial-distribution probabilities.
- I know the Poisson approximation formula for  $\text{binomial}(n, p)$  where  $n$  is large and  $p$  is small.
- I know the  $\text{Poisson}(\lambda)$  distribution.
- I can use the Poisson distributions model counting rare events.
- I know the formula for the  $\text{geometric}(p)$  distribution.
- I know how to match the story of a “waiting time” random variable with the  $\text{geometric}(p)$  distribution.
- I know the shortcut formulas for expected value and variance of random variables with the  $\text{binomial}(n, p)$ ,  $\text{Poisson}(\lambda)$ , and  $\text{geometric}(p)$  distributions.
- I can solve “urn problems,” matching chance experiments with the story of drawing balls from an urn with a certain number of red balls, using the random variable that counts the number of red balls drawn.

## Counting

- I can count using the multiplication rule when counting done in stages with “homogeneity” in the possibility tree at each stage.
- I can count by matching a counting situation with the stories of permutations and combinations.
- I can use Durrett’s terminology and notation for permutations and combinations.
- I can count using the multinomial counting method.
- I can combine counting methods to solve complex problems, such as splitting a class into groups where the order does not matter or making license plates where letters and numbers do not have fixed positions.

- I can use these counting methods to calculate probabilities in situations where all outcomes are equally likely.